CSA1737 - ARTIFICIAL INTELLIGENCE

ASSESSMENT TOOLS [CO1 - AT1]

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1) Intelligent Agents in a Smart Home Automation System:

Intelligent Agent is a System that perceives its environment through Sensors takes action through actuators to achieve a particular objective. In Smart Home Automation System sensors collect information such as motion, temperature, light intensity & time of day. System uses this information to control lighting, air conditioning, security device automatically.

four types of intelligent agents:

1: Simple Reflex Agent: It acts based on current inputs using condition-action rules. for eg: it switches on light when motion is detected. It's fast and simple but cannot remember past events.

2: Model Based Agent: It maintains an internal state using current & previous information. for eg: it remembers room occupancy & control lights accordingly. It handles changing environments but requires memory.

3: Goal-Based Agent: It select options actions to achieve specific goals. for eg: it locks door & switches off unecessary appliances when everyone leaves home. It supports planning but

requires more computation.

4: Utility-Based Agent: It select actions that provide the greatest overall benefit. For example, it balances comfortable temperature, electricity consumption & security. It handles competing objectives but requires a suitable utility function.

Simple Reflex agents respond to current Inputs, model based agents maintain state, goal-based agents work toward objectives, utility-based agents choose the most beneficial actions. A hybrid model-based agent with goal-based planning is most suitable. It remembers environmental conditions, plans actions & Balances comfort, energy efficiency, security. Simple reflex rules can also provide immediate responses to emergencies. A Hybrid Intelligent agent provides effective smart home automation by combining quick responses, memory, planning and efficient decision making.

2) Uniformed Search Strategies for a Robot in Unknown Maze:

Uninformed search explores possible paths without prior knowledge of the maze layout. Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help robot to find exit.

1: Uniform Cost Search (UCS): expands the position with the lowest total path cost. The robot explores neighboring positions &

continues until it reaches the exit.

* Adv: finds the least-cost path when step cost are positive, handles different movement costs.

* Disadv: Requires more memory & explore many positions

* Time & Space: Exponential in the worst case.

2. Iterative Deepening Search (IDS): IDS repeatedly performs depth-limited, depth-first search, increasing the depth limit until the exist is found.

* Adv: Uses less memory & find the shallowest solution when all movements have equal costs.

* Disadv: Repeats searches at smaller depths and is not optimal for different movement costs.

* Time: $O(b^d)$ Space: $O(bd)$ where b is branching factor & d is solution depth.

Relationship with intelligent agents:

→ Simple reflex Agent: Responds to immediate sensor inputs but may repeat paths.

→ Model Based agent: Remembers visited positions and discovered walls.

→ Goal Based agent: Uses search to reach the exit.

→ Utility Based agent: Select paths based on cost, distance & risk.

A model-based, Goal-based agent using IDS is suitable when memory is limited and all movements have equal costs. If movement differs, UCS is preferable because it finds the least-cost path. The robot must update its internal map as it discovers the maze.

UCS is suitable for minimizing path cost, while IDS is memory-efficient. Combining an appropriate search strategy with a model based, Goal Based agent improves the robot's decision making & ability to reach the exit.